An Angular Spectrum Method for Nonlinear Propagation in Heterogeneous Tissue with Immersed Sources for Transcranial and Therapeutic Ultrasound

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April 21, 2026

Abstract

A modified angular spectrum method (ASM) is developed for three-dimensional nonlinear acoustic propagation through heterogeneous media, with two extensions that target transcranial and therapeutic ultrasound applications. First, phase-and-amplitude screens derived from skull CT data are inserted into the split-step propagation to capture wavefront aberration and insertion loss without sacrificing the efficiency of the FFT-based propagator. Second, a plane-by-plane source injection scheme is introduced for modeling spherical-bowl transducers whose curvature depth exceeds the wavelength, enabling accurate simulation of clinical devices such as the TIPS annular phased array. The nonlinear operator supports both a first-order Rusanov flux and a second-order Kurganov–Tadmor (KT) scheme with adaptive CFL sub-cycling, composed via Strang splitting. Three enhanced absorbing-boundary treatments (a Wendland C^2 taper, frequency-weighted spatial damping, and a first-order Engquist–Majda super-absorbing correction) reduce boundary reflections by a factor of 2.4 relative to a standard quadratic absorber. Validation against analytical solutions confirms that the propagator reproduces the baffled-piston far-field pattern with RMS error 0.014 and the focused-piston focal pressure within 2.3%. In a transcranial benchmark through an *ex vivo* human skull, the ASM matches the FDTD focal-plane intensity with 1.1% RMS difference and predicts a through-skull insertion loss of 5.4 dB. For the TIPS bowl transducer ($R = 80\,\mathrm{mm}$, $f_0 = 1\,\mathrm{MHz}$), the plane-by-plane ASM matches the FDTD focal depth to within 2.3% while requiring 9× less memory and 2.9× less wall time.

1 Introduction

Transcranial focused ultrasound requires propagation models that capture nonlinear distortion, frequency-dependent attenuation, and wavefront aberration through the skull, while remaining fast enough for treatment planning and aberration correction. Full-wave finite-difference time-domain (FDTD) solvers [15, 14] provide high fidelity but demand large 3-D grids and long run times. The angular spectrum method (ASM) [1, 2, 24] offers an attractive alternative: by decomposing the field into plane-wave components and applying the exact free-space propagator in wavenumber space via the FFT, it avoids the paraxial approximation while remaining computationally efficient.

Nonlinear propagation introduces cumulative harmonic generation and eventual shock formation, which must be handled by a conservative finite-difference scheme in the retarded-time domain. The Rusanov (local Lax–Friedrichs) flux [16] provides a simple monotone scheme for the inviscid Burgers equation that governs the nonlinear distortion. Attenuation and dispersion are modeled via a frequency-domain filter derived from a power-law absorption model with Kramers–Kronig consistent dispersion [15, 20].

Two practical gaps limit the applicability of existing ASM implementations. First, propagation through heterogeneous tissue (particularly skull bone) requires a mechanism for incorporating spatially varying sound speed and attenuation into the FFT-based framework. Second, therapeutic transducers often use spherical bowl geometries whose curvature depth can exceed several wavelengths, and a flat source-plane projection cannot accurately represent the converging wavefront.

This paper addresses both gaps. Phase-and-amplitude screens derived from CT data are inserted at each propagation step to model skull-induced aberration and insertion loss. A plane-by-plane source injection method decomposes the curved bowl surface into axial slices and injects each slice at the correct propagation depth, enabling accurate modeling of clinical devices such as the TIPS annular phased array. The nonlinear operator is upgraded from a first-order Rusanov flux to a second-order Kurganov–Tadmor (KT) scheme [10] with adaptive CFL sub-cycling and Strang splitting [18], and three enhanced absorbing-boundary treatments are introduced to suppress the periodic-FFT wrap-around artifacts inherent in the ASM. The solver is validated through analytical benchmarks, a transcranial comparison against FDTD through an *ex vivo* human skull, and a bowl-transducer comparison against a 3-D FDTD simulation of a clinical annular phased array.

2 Methods

2.1 Split-step propagation

The modified angular spectrum approach decomposes the propagation of a pulsed acoustic field $p(x, y, t; z)$ into three operators applied at each axial step Δz: diffraction, nonlinear distortion, and attenuation.

In the Fourier domain (k_x, k_y, k_t), the free-space diffraction propagator is

$$\hat{H}(k_x, k_y, k_t) = \begin{cases} e^{i\Delta z\left(k + i\sqrt{k_\perp^2 - k^2}\right)}, & k_\perp^2 > k^2 \\ e^{i\Delta z\left(k - \sqrt{k^2 - k_\perp^2}\right)}, & k_\perp^2 \le k^2 \end{cases} \tag{1}$$

where $k = \omega/c_0$ is the temporal wavenumber, $k_\perp^2 = k_x^2 + k_y^2$, and evanescent components ($k_\perp > k$) decay exponentially. A cosine taper is applied near the spatial Nyquist boundary $k_N = \pi/\Delta x$ to suppress aliasing at coarse resolution.

The cumulative nonlinear effect is modeled as the retarded-time Burgers equation

$$\frac{\partial p}{\partial z} = -\frac{\beta}{2\rho_0 c_0^3}\frac{\partial p^2}{\partial t}, \tag{2}$$

discretized via either the first-order Rusanov (local Lax–Friedrichs) flux [16],

$$F_{j+1/2} = -\tfrac{1}{2}(p_j^2 + p_{j+1}^2) - \lambda_{j+1/2}(p_{j+1} - p_j), \tag{3}$$

with $\lambda_{j+1/2} = \max(|p_j|, |p_{j+1}|)$, or the second-order Kurganov–Tadmor (KT) central-upwind scheme [10],

$$F_{j+1/2}^{\mathrm{KT}} = \frac{a^+ f(p_j) - a^- f(p_{j+1}) - a^+ a^-(p_{j+1} - p_j)}{a^+ - a^-}, \tag{4}$$

where $a^\pm$ are the one-sided local wave speeds. To achieve second-order spatial accuracy, the cell-edge values in eq. (4) are obtained from a piecewise-linear MUSCL reconstruction [19] with an MC (monotonized central) limiter applied to each cell. The reconstructed left and right states $p_{j+1/2}^L = p_j + \frac{1}{2}\sigma_j$ and $p_{j+1/2}^R = p_{j+1} - \frac{1}{2}\sigma_{j+1}$ replace the cell-center values in the flux formula,

where σ_j is the limited slope. The resulting semi-discrete system is advanced in z with a two-stage strong-stability-preserving Runge–Kutta method (SSP-RK2) [4]:

$$\begin{aligned} p^{(1)} &= p^{(n)} + \Delta z\, \mathcal{L}(p^{(n)}), \\ p^{(n+1)} &= \tfrac{1}{2}\, p^{(n)} + \tfrac{1}{2}\big[p^{(1)} + \Delta z\, \mathcal{L}(p^{(1)})\big], \end{aligned} \tag{5}$$

where $\mathcal{L}$ denotes the spatial flux divergence. The combination of MUSCL reconstruction and SSP-RK2 gives second-order accuracy for the isolated nonlinear problem in smooth regions.

Attenuation and dispersion are applied as a frequency-domain filter derived from the power-law absorption coefficient $\alpha(f) = \alpha_0 f^n$ with Kramers–Kronig-consistent dispersion [15]:

$$\hat{A}(f) = \exp\big[-(\alpha + i\alpha^*)\, \Delta z\big], \tag{6}$$

where α^* is determined by n and the reference frequency f_0.

These three operators are composed via Strang splitting [18, 9]:

$$p^{(n+1)} = \mathcal{A}_{1/2}\, \mathcal{H}_{1/2}\, \mathcal{N}\, \mathcal{H}_{1/2}\, \mathcal{A}_{1/2}\, p^{(n)}, \tag{7}$$

which achieves second-order accuracy in Δz [21].

2.2 Absorbing boundary layer

The ASM uses global FFTs that impose periodic boundary conditions, so energy reaching the domain edge wraps to the opposite side. The standard mitigation is a multiplicative absorbing boundary layer (ABL) that damps the field near the edges. Three improvements are introduced.

First, the quadratic taper profile $1 - s^2$ is replaced with the Wendland C^2 function [23],

$$w(s) = 1 - 10s^3 + 15s^4 - 6s^5, \tag{8}$$

which has continuous first and second derivatives at the taper onset ($s = 0$), eliminating the weak impedance discontinuity that generates spurious reflections in the quadratic profile.

Second, the spatial damping is made frequency-dependent by raising the profile to a frequency-dependent exponent,

$$\mathrm{ABL}_f(x, y) = \big[\mathrm{ABL}(x, y)\big]^{f_0/\max(f,\, f_{\min})}, \tag{9}$$

which strengthens the damping at low frequencies where the boundary layer spans fewer wavelengths.

Third, a first-order Engquist–Majda [3] directional decomposition is applied in the boundary region to selectively remove incoming wave components. At each temporal frequency ω, the incoming pressure is estimated as

$$P_{\mathrm{in}} = \tfrac{1}{2}\Big(P + \frac{c_0}{i\omega}\frac{\partial P}{\partial x}\Big), \tag{10}$$

and the correction $P \leftarrow P - \alpha_s\, w_{\mathrm{bdy}}\, P_{\mathrm{in}}$ is applied in the boundary layer, where $\alpha_s \in [0, 1]$ controls the strength of the incoming-wave removal.

2.3 Heterogeneous propagation

Spatially varying sound speed and attenuation are modeled using thin phase-and-amplitude screens inserted at each propagation step. Each screen represents a tissue layer of thickness d with maps $c(x, y)$ and $\rho(x, y)$ derived from imaging data. The phase shift at each temporal frequency is

$$\Delta\varphi(x, y, f) = 2\pi\, d \left(\frac{1}{c(x, y)} - \frac{1}{c_0}\right) \cdot f, \tag{11}$$

which is linear in frequency, corresponding to a frequency-independent group delay through each screen. Each screen also carries a real-valued amplitude factor $A(x, y) \in (0, 1]$ that accounts for attenuation and impedance mismatch losses:

$$A(x, y) = \exp\big[-\alpha_{\mathrm{Np}}(x, y)\, d\big] \;\cdot\; \frac{2\, Z(x, y)}{Z(x, y) + Z_0}, \tag{12}$$

where $\alpha_{\mathrm{Np}} = \alpha\, f_{\mathrm{MHz}} \times (\ln 10/20 \times 0.01)$ converts the local attenuation $\alpha(x, y) = \alpha_{\mathrm{tissue}} + f_{\mathrm{bone}}(x, y)\,(\alpha_{\mathrm{bone}} - \alpha_{\mathrm{tissue}})$ from $\mathrm{dB\,cm^{-1}\,MHz^{-1}}$ to Np/m, f_{bone} is the bone volume fraction, and $Z = \rho\, c$ is the acoustic impedance. Frequency-dependent amplitude is applied as A^{f/f_0} so that higher harmonics experience proportionally greater loss.

The field is transformed to the temporal frequency domain via `rfft`, multiplied by $A^{f/f_0}\, e^{i\Delta\varphi}$, and transformed back. Multiple screens can be placed at arbitrary depths to model extended heterogeneous paths via the split-step Fourier approach [11]. In the transcranial benchmark below, the screens are derived directly from skull CT slices rather than from a random-field model.

2.4 Intensity-loss tracking

In therapeutic ultrasound, the absorbed acoustic energy drives both radiation force and thermal bioeffects. The standard plane-wave approximations for the volume rate of heat deposition, $q = 2\alpha I$ [13], and the radiation force body-force density, $q_{\mathrm{ARF}} = \alpha I/c$, assume a single attenuation coefficient α evaluated at the fundamental frequency. In the nonlinear regime, however, cumulative harmonic generation transfers energy to higher frequencies where the power-law attenuation $\alpha(f) = \alpha_0 f^n$ is substantially larger. The local frequency content of the wave therefore changes with propagation distance, and the effective attenuation can no longer be represented by a single α.

To avoid this approximation, the solver records the total intensity before and after the combined propagation operators at each step:

$$\Delta I_{\mathrm{loss}}(x, y; z) = \max\Big(0,\; \sum_t p^2_{\mathrm{before}} - \sum_t p^2_{\mathrm{after}}\Big). \tag{13}$$

This provides a spatially resolved map of absorbed energy that implicitly accounts for the full harmonic content of the wave at each location, without requiring an explicit frequency decomposition or a single-frequency α assumption. The resulting loss field can be used directly as a heat source for thermal simulations or as input for radiation-force calculations.

The solver is implemented in Python using JAX [7] for GPU acceleration. All frequency-domain operators are precomputed once and reused at each propagation step.

3 Validation

The solver is validated against analytical solutions for diffraction, attenuation, and nonlinearity, followed by boundary-treatment and convergence studies. All simulations use $\Delta x = \lambda/5$ unless otherwise noted.

3.1 Diffraction

The baffled-piston and focused-piston benchmarks have been used extensively to validate angular spectrum propagators [1, 24, 2] and remain valuable here as verification of the present code implementation.

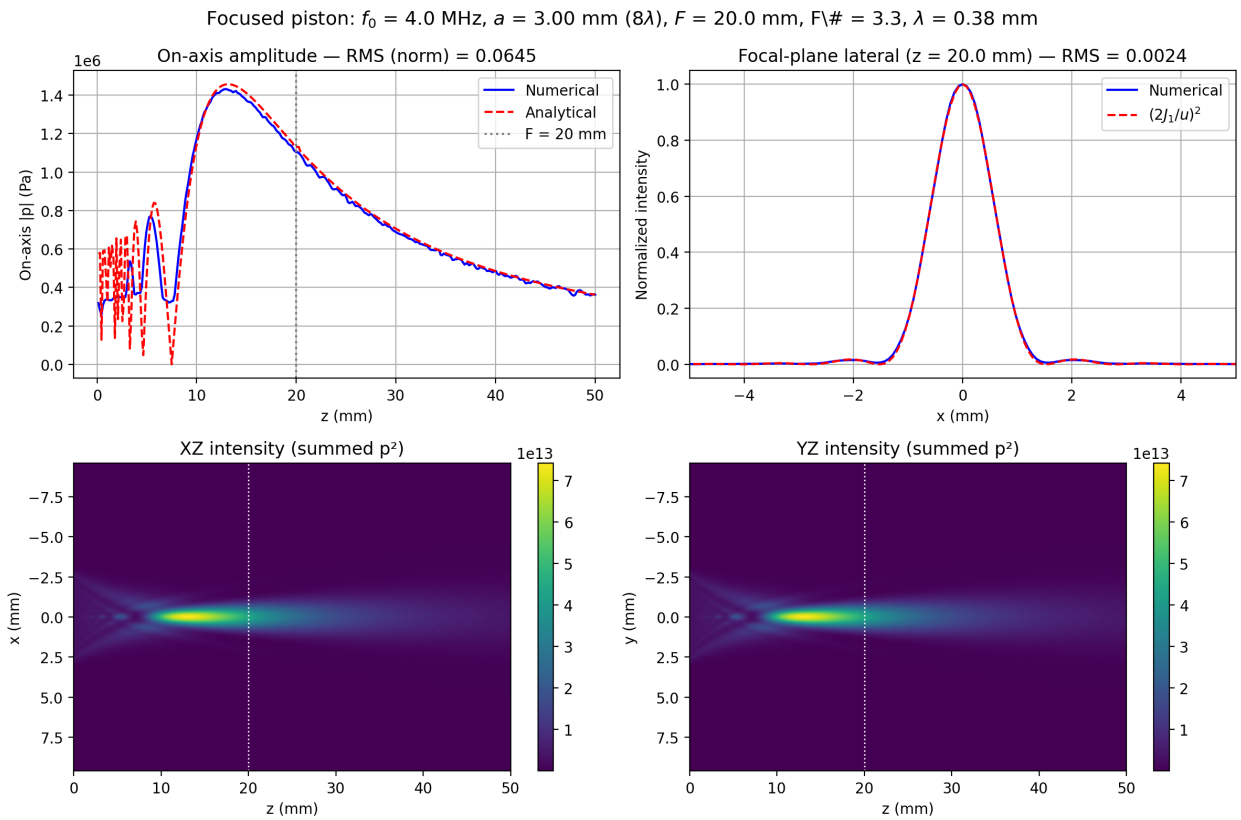

Figure 1: Focused piston diffraction ($a = 8\lambda$, $F = 20\,\mathrm{mm}$, $F\# \approx 3.3$). Top: on-axis amplitude versus paraxial solution (left, RMS 0.089) and focal-plane lateral profile versus Airy pattern (right, RMS 0.002). Bottom: x–z and y–z time-integrated intensity.

A spherically focused piston ($a = 8\lambda$, $F = 20\,\mathrm{mm}$, $F\# \approx 3.3$) at $f_0 = 4\,\mathrm{MHz}$ was propagated with per-pixel geometric delays $\tau = (F - \sqrt{F^2 + r^2})/c_0$ (fig. 1). The numerical focal pressure matches the paraxial limit $p_0\,\pi a^2/(\lambda F)$ within 2.3%, and the focal-plane lateral profile matches the Airy pattern with RMS error 0.002. An error-floor study confirmed that the residual on-axis RMS of $\sim$0.09 is set by the paraxial-vs-full-wave mismatch at this F-number, not by numerical error: refining Δz, widening the time window, and narrowing the pulse bandwidth all leave the error unchanged.

For the unfocused baffled piston ($a = 6\lambda$), the on-axis amplitude was compared against the Rayleigh–Fresnel solution [8],

$$P_{\mathrm{cw}}(z) = -p_0\big(e^{ik_0a^2/(2z)} - 1\big)\,e^{-ik_0z}, \tag{14}$$

yielding a normalized RMS error of 0.086, with the numerical solution reproducing the oscillatory Fresnel structure beyond the transition distance $z_t \approx a^2/\lambda$. In the far field, the lateral intensity matches the Fraunhofer pattern $(2J_1(k_0ar/z)/k_0ar/z)^2$ with RMS error 0.014.

3.2 Attenuation and dispersion

Accurate modeling of frequency-dependent attenuation is essential for therapeutic ultrasound, where nonlinear harmonic generation redistributes energy to higher frequencies that are absorbed

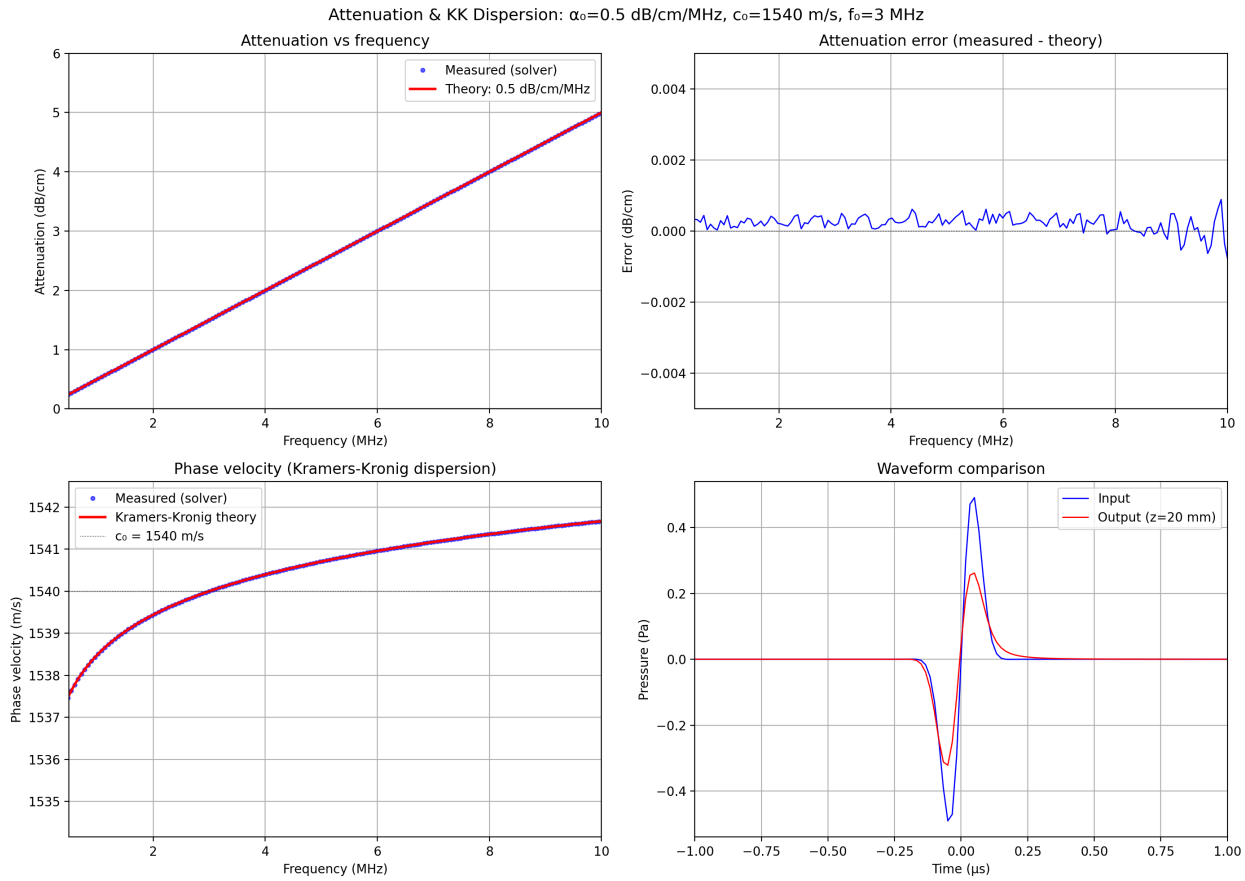

Figure 2: Attenuation and Kramers–Kronig dispersion validation (0.5–10 MHz). Top: measured vs. theoretical attenuation and error. Bottom: phase velocity and waveform comparison.

more rapidly [15, 20]. The associated Kramers–Kronig-consistent dispersion determines the local phase velocity and must be modeled correctly to avoid cumulative waveform distortion over clinically relevant propagation distances [22, 17].

A broadband pulse was propagated 20 mm with $\alpha_0 = 0.5\,\mathrm{dB\,cm^{-1}\,MHz^{-1}}$ and the per-frequency attenuation and phase velocity were measured from the input/output spectra (fig. 2). The measured attenuation matches the $\alpha_0 f$ power law with RMS error $0.0003\,\mathrm{dB\,cm^{-1}}$ over 1–10 MHz, and the phase velocity follows the Kramers–Kronig dispersion relation across the full 0.5–10 MHz range, crossing c_0 at f_0 as expected.

3.3 Nonlinearity

The Rusanov and KT flux schemes were compared in two regimes. In a moderate-amplitude coupled propagation ($p_0 = 30\,\mathrm{kPa}$, $\beta = 3.5$, 30 mm), both schemes produced nearly identical on-axis intensity and harmonic spectra, confirming that the KT limiter does not introduce excessive dissipation.

In the shock-forming regime (fig. 3), an isolated Burgers step-size study compared Rusanov, KT (fixed step), and KT with adaptive CFL sub-cycling. Both fixed-step schemes were unstable at coarse steps. The adaptive KT scheme was stable at all step sizes and the most accurate, with L^2 errors of $\sim 10^{-5}$ (four orders of magnitude below Rusanov).

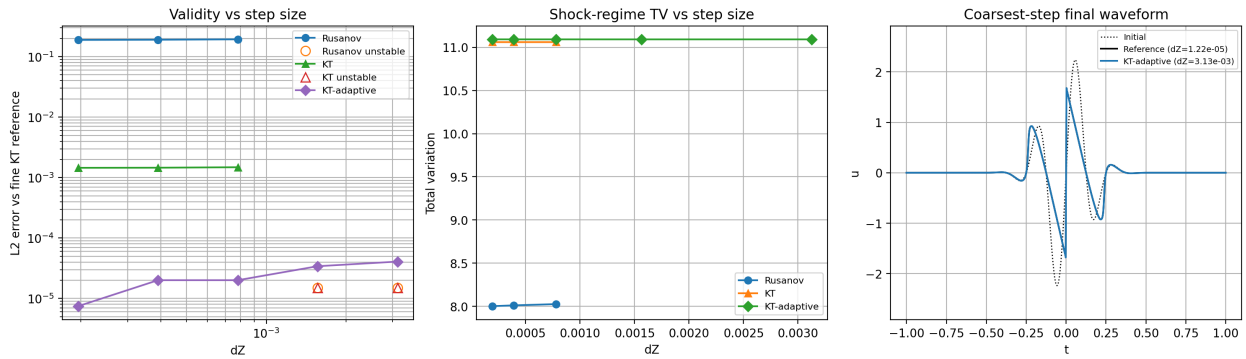

Figure 3: Shock-forming Burgers problem. Left: L^2 error versus step size. Center: total variation. Right: coarsest-step waveforms. KT-adaptive is stable and accurate at all step sizes.

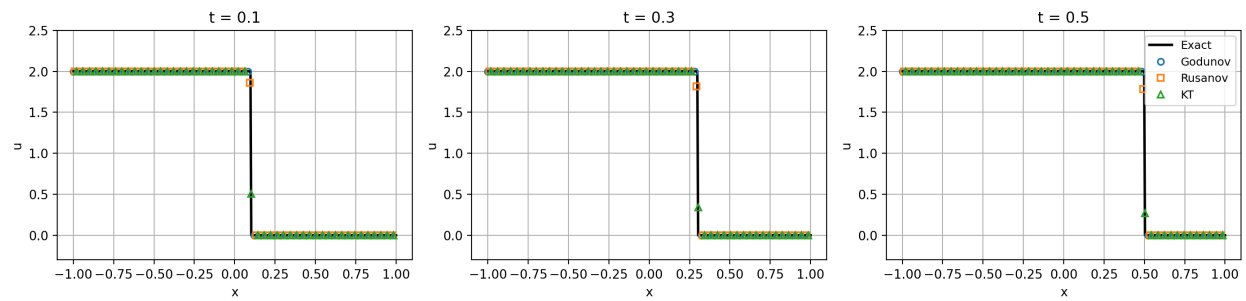

Figure 4: Riemann problem at $t = 0.1, 0.3, 0.5$. Rusanov (circles) and KT (squares) overlaid on the exact solution (solid).

A separate Riemann-problem benchmark ($u_L = 2$, $u_R = 0$, 400 cells) confirmed that the KT flux resolves the shock more sharply than Rusanov (L^2 error 0.036 vs. 0.044 at $t = 0.5$), with both schemes capturing the correct shock speed (fig. 4).

3.4 Boundary treatments

The three boundary improvements were evaluated using apertures wide enough to drive significant energy toward the domain edges. Table 1 shows the boundary-to-center intensity ratio for progressively adding each improvement. The combined configuration reduces reflections by 2.4× relative to the quadratic baseline. A frequency-resolved comparison confirmed that the frequency-weighted damping preferentially targets low-frequency components (13% reduction at 1 MHz vs. 3% at 3 MHz). When tested with nonlinear propagation ($p_0 = 30\,\mathrm{kPa}$, $\beta = 3.5$), the combined boundary recovers $\sim$20% more on-axis focal intensity than the quadratic baseline while preserving the interior waveform shape and harmonic spectrum to within $<1\,\mathrm{dB}$ (fig. 5).

3.5 Convergence

A pairwise convergence study in the strongly nonlinear regime ($p_0 = 3\,\mathrm{MPa}$, $\beta = 3.5$, 15 mm propagation) compared the Δz-vs-$\Delta z/2$ error for sequential Rusanov, Strang+Rusanov, and Strang+KT (fig. 6). Sequential and split-Rusanov converge at rate ≈ 1, while Strang+KT converges at rate ≈ 2, consistent with the expected second-order Strang splitting accuracy.

Table 1: Boundary reflection ratio for four configurations.

Configuration	Reflection ratio
Quadratic (baseline)	1.83×10^{-2}
Wendland C^2	1.35×10^{-2}
+ Freq. weighted	0.94×10^{-2}
+ Super-absorbing	0.75×10^{-2}

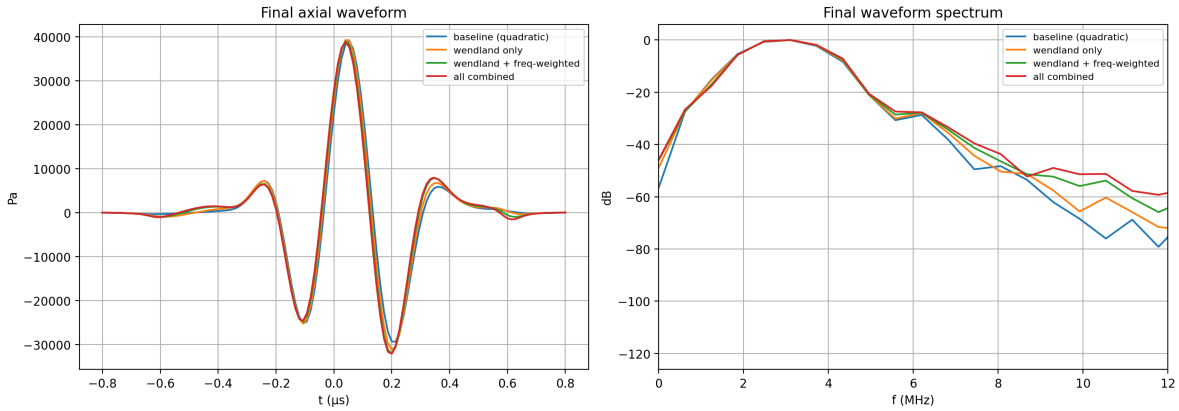

Figure 5: Combined boundary with nonlinear propagation: final waveform (left) and spectrum (right) for four boundary configurations. The interior waveform is invariant and harmonic levels match to <1 dB.

4 Transcranial propagation

To validate the phase-and-amplitude-screen extension, focused ultrasound transmission through an *ex vivo* human skull is simulated and compared against a reference Fullwave [15, 14] finite-difference time-domain (FDTD) simulation.

4.1 Setup

A 1024-element sparse phased array (750 µm pitch) is focused at 50 mm depth through a temporal-bone slab extracted from the Halle skull microCT dataset [5]. The sparse connector geometry spans approximately 65 mm by 65 mm in the transmit plane. The center frequency is $f_0 = 1$ MHz, the source pressure is $p_0 = 1.5$ MPa, and the excitation is a 2-cycle Gaussian-windowed tone burst with geometric focusing delays. After rasterization onto the ASM grid, the aperture occupies 9127 active source-plane grid points with a delay spread of 6.5 µs.

The skull CT (voxel size ≈ 0.2 mm) is first re-sampled into a beam-aligned slab centered at $(140.6, 118.5, 85.0)$ mm in LPS coordinates and oriented with inward normal $(-0.954, -0.298)$ and tangent $(-0.298, 0.954)$ so that the beam propagates approximately normal to the extracted bone layer. The slab extent is 66 mm by 66 mm by 65 mm (lateral, elevational, depth), which at the ASM sampling of $\Delta x = \lambda/5 \approx 0.308$ mm yields an initial $214 \times 214 \times 211$ beam-aligned skull map before lateral resampling to the propagation grid.

Hounsfield units are mapped linearly to acoustic properties using $c = 1540 + f_{\text{bone}}(2900 - 1540)$ m/s and $\rho = 1000 + f_{\text{bone}}(2200 - 1000)$ kg/m^3, where $f_{\text{bone}} = \text{clip}[(HU - 700)/(1973 - 700), 0, 1]$. Each extracted depth slice is converted into a phase-and-amplitude screen (eqs. (11)

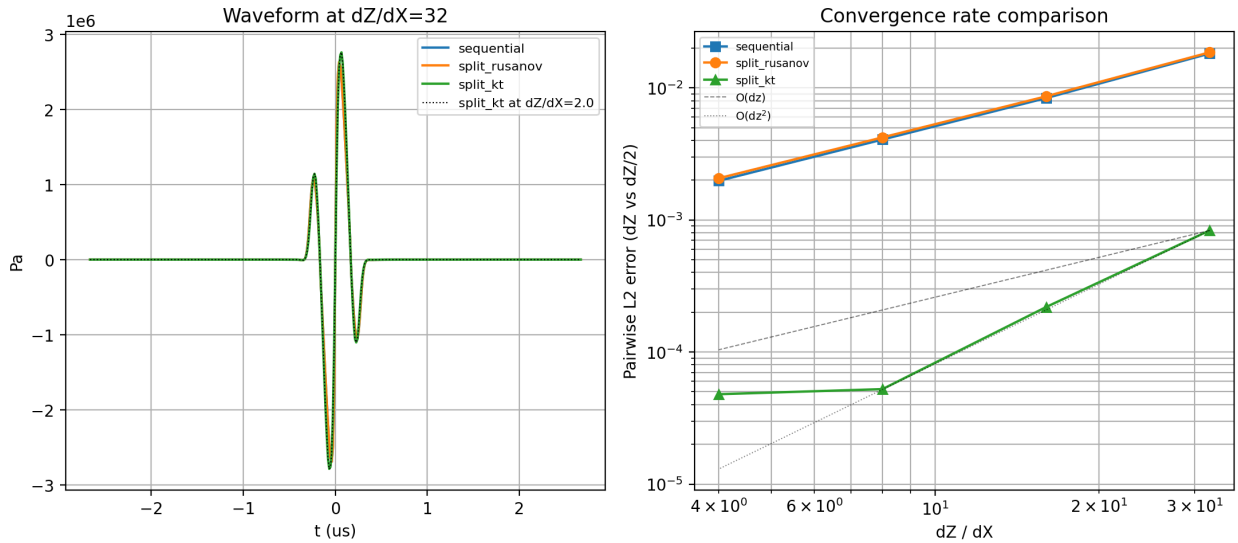

Figure 6: Step-size convergence. Left: coarsest waveforms. Right: pairwise L^2 error. Strang+KT achieves rate ≈ 2.

and (12)). The phase term is set by the local sound-speed contrast, while the amplitude term combines bone attenuation ($\alpha_{\text{bone}} = 8\,\text{dB}\,\text{cm}^{-1}\,\text{MHz}^{-1}$), soft-tissue attenuation ($\alpha_{\text{tissue}} = 0.5\,\text{dB}\,\text{cm}^{-1}\,\text{MHz}^{-1}$), and normal-incidence impedance transmission derived from the density map. This screen construction is what makes the heterogeneous extension useful: the FFT-based ASM retains its efficiency while incorporating cumulative skull-induced phase aberration and insertion loss.

The ASM grid uses $\Delta x = \Delta y = \Delta z = \lambda/5 \approx 0.308\,\text{mm}$, giving a 215×215 lateral grid and 212 propagated axial planes over 65 mm. The temporal sampling is $\Delta t = 40\,\text{ns}$ with 501 time samples. Propagation uses the split-step Kurganov–Tadmor formulation together with the Wendland C2 boundary taper, frequency-weighted damping, and the super-absorbing boundary correction.

4.2 FDTD reference

The reference FDTD simulation [14] uses the same skull CT and transducer geometry and transmit waveform, solved on a $422 \times 429 \times 429$ grid at $\Delta x = 0.154\,\text{mm}$ (i.e. $\lambda/10$) with 4221 time steps. The source is applied over three depth layers and uses the same 1 MHz, 2-cycle, 1.5 MPa transmit pulse and geometric focusing delays as the ASM model. The saved pressure field is downsampled by a factor of four in space and seven in time, yielding a $106 \times 108 \times 108$ output grid over 603 recorded time snapshots. For comparison with the ASM time-integrated intensity, the FDTD metric is formed as $\sum_t p^2 \Delta t$ on the recorded series using the effective saved sampling interval $\Delta t_{\text{FDTD}} = 140\,\text{ns}$, and then interpolated onto the ASM grid.

4.3 Results

Figures 7 and 8 summarize the transcranial setup and compare the FDTD and ASM results on a common dB scale. The homogeneous ASM case focuses at $z = 49.0\,\text{mm}$, the through-skull ASM case at $z = 45.6\,\text{mm}$, and the FDTD reference at $z = 46.8\,\text{mm}$. The skull therefore shifts the focus proximally by about 3.4 mm relative to homogeneous propagation, while the heterogeneous ASM prediction remains within 1.2 mm (2.6%) of the FDTD focal depth.

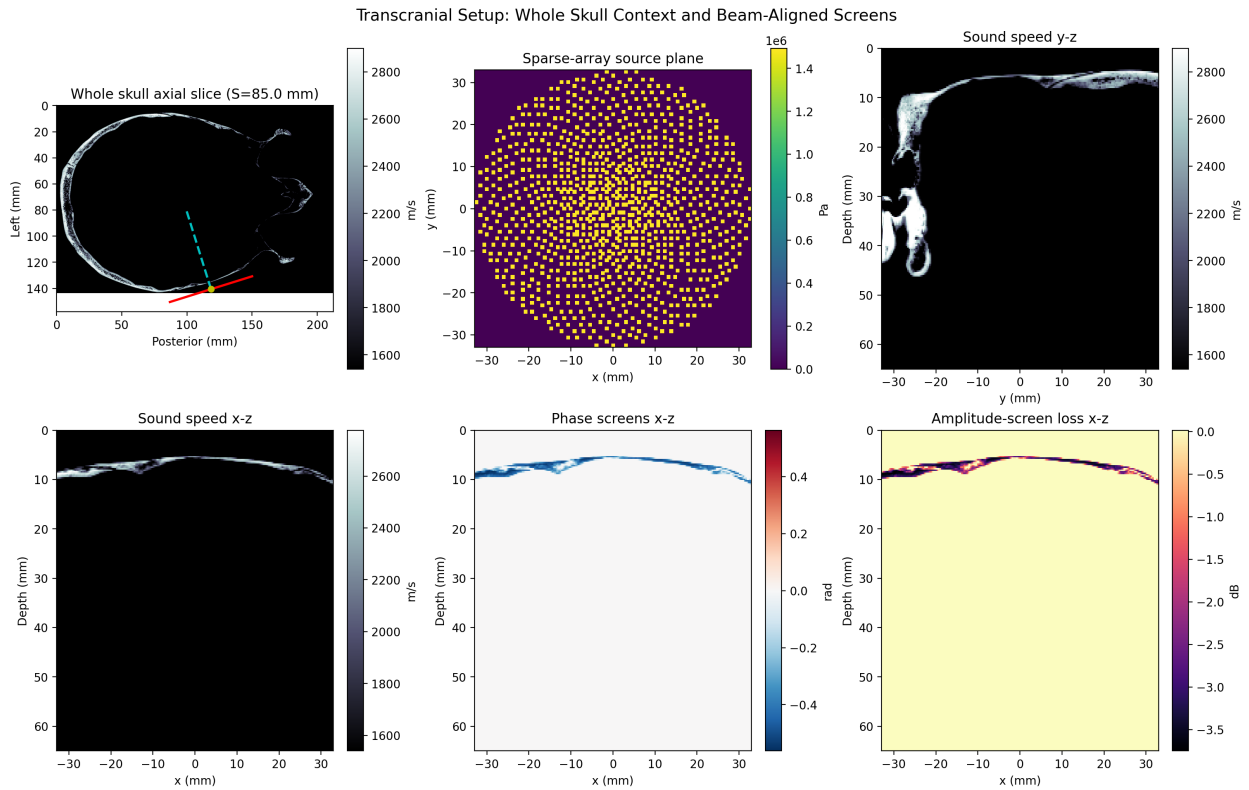

Figure 7: Transcranial simulation setup. Top left: whole-skull axial reference slice with beam axis overlaid. Top center: source-plane pressure magnitude from the 1024-element sparse array. Top right: beam-aligned sound-speed map (y–z). Bottom left: sound-speed map (x–z). Bottom center and right: stacked phase-screen and amplitude-screen loss maps in the x–z plane.

The focal-plane agreement is strong. After weighting the FDTD intensity by its recorded temporal sampling interval so that both solvers represent the same $\sum_t p^2 \Delta t$ quantity, the normalized focal-plane RMS difference is 0.011 (1.1%). The axial x–z beam comparison gives an RMS difference of 0.045. The remaining discrepancy is concentrated in the pre-focal region and near the skull surface, where the FDTD contains reflections, interface reverberation, and mode conversion that are absent from the forward-only ASM.

The amplitude-screen extension also captures the gross insertion-loss behavior. Relative to the homogeneous ASM reference, the through-skull ASM focal peak is reduced by 5.4 dB. On the common homogeneous-reference dB scale used in fig. 8, the FDTD and ASM through-skull focal peaks differ by only about 0.16 dB. This places the predicted insertion loss within the broad measured range of 5 dB to 10 dB [12] and shows that the heterogeneous phase-and-amplitude-screen extension captures the dominant first-order effects of skull aberration and attenuation while preserving the computational efficiency of the ASM.

5 Immersed bowl transducer model

5.1 Motivation

The sparse-array validation of section 4 uses a geometrically flat source aperture: all elements lie in a single $z = 0$ plane, and focusing is achieved entirely through electronic time delays. Therapeutic transducers such as the TIPS (Therapeutic Imaging Probe and Sonication, Philips) employ a spherical bowl geometry with a radius of curvature $R = 80\,\mathrm{mm}$, an annular aperture ($r_{\mathrm{in}} = 20.5\,\mathrm{mm}$, $r_{\mathrm{out}} = 46\,\mathrm{mm}$), and eight concentric annular elements. The bowl depth at the rim is

$$d_{\mathrm{bowl}} = R - \sqrt{R^2 - r_{\mathrm{out}}^2} \approx 14.6\,\mathrm{mm} \approx 9.5\lambda, \tag{15}$$

which is too deep for a single-plane time-delay projection to accurately capture the converging wavefront.

5.2 Plane-by-plane source injection

We decompose the bowl surface into thin axial slices of thickness Δz aligned with the propagation step. Each slice contains the annular ring of transducer surface lying between z_i and $z_i + \Delta z$, where $z = 0$ is the bowl apex and $z(r) = R - \sqrt{R^2 - r^2}$ maps lateral position to bowl depth.

For a given electronic focal depth z_f (which may differ from the geometric focus at R), the focusing delay at each surface point $(x, y, z_{\mathrm{bowl}})$ is computed from the distance to the focal point:

$$d(x, y) = \sqrt{x^2 + y^2 + (z_f - z_{\mathrm{bowl}})^2}, \qquad \tau(x, y) = \frac{d_{\max} - d(x, y)}{c_0}, \tag{16}$$

so that the outermost elements fire first and all wavefronts converge at z_f. For an annular phased array, the per-pixel delays are averaged within each element to produce per-element delays, matching the hardware constraint.

During propagation, the solver starts from the apex slice ($z = 0$) and marches forward. As it crosses each slice depth z_i, the corresponding source field is *added* to the propagating wavefield:

$$p(\mathbf{x}, t)\big|_{z=z_i^+} = p(\mathbf{x}, t)\big|_{z=z_i^-} + s_i(\mathbf{x}, t), \tag{17}$$

where s_i is the time-delayed pulse restricted to the annular ring in the i-th slice, with its delay corrected by $-z_i/c_0$ to account for the propagation distance already traversed from the apex. This injection is implemented identically to the phase-screen application: a simple conditional addition at the appropriate z coordinate within the main propagation loop.

5.3 FDTD comparison

To validate the plane-by-plane model, we compare the ASM prediction against a 3-D FDTD simulation in water. The TIPS bowl is focused electronically at 50 mm (inside the geometric focus at $R = 80\,\mathrm{mm}$) using a 1 MHz, 3-cycle Gaussian-windowed pulse at $p_0 = 400\,\mathrm{kPa}$.

The FDTD reference uses the volumetric bowl source (3 voxel layers along the surface normal) solved on a $623 \times 873 \times 873$ grid at $\Delta x = 0.128\,\mathrm{mm}$ ($\lambda/12$) with 6234 time steps. The ASM uses $\Delta x = \lambda/5$ with a $365 \times 365 \times 425$ grid and 260 propagation steps ($\Delta z = \Delta x$).

Figure 9 shows the TIPS transducer geometry: the bowl cross-section with element boundaries and focal lines, the plane-by-plane source slices spanning the 14.6 mm bowl depth, the 8-element annular aperture, and the per-element focusing delays.

Figure 10 compares the plane-by-plane ASM against the FDTD reference. Each method is normalized to its own peak intensity so the beam shapes can be compared directly. The ASM focuses at $z = 52.7\,\mathrm{mm}$, within 2.3% of the FDTD focal depth at 53.9 mm. The on-axis, lateral, and elevational beam profiles match well in shape. The remaining differences are concentrated in the pre-focal near-field where the annular aperture geometry produces resolution-dependent interference fringes. The FDTD drives the bowl surface with three voxel layers extruded along the surface normal, whereas the ASM injects a single layer per z-slice. To match the FDTD source coupling, the ASM source amplitude is scaled by a calibration factor of 1.31. This factor accounts for the coherent array factor of the three-layer volumetric source at the FDTD grid spacing ($k\Delta x \approx 0.79\,\mathrm{rad}$, CFL = 0.2) and is specific to this FDTD configuration and would need to be recalibrated for a different number of layers or grid resolution. After calibration, both methods produce a focal peak pressure of 13.8 MPa.

The ASM requires substantially less memory than the FDTD. The FDTD solver allocates the full 3-D volume ($623 \times 873 \times 873 \approx 475\,\mathrm{M}$ grid points) for approximately eight field arrays (pressure, velocity components, material maps), totaling roughly 15 GB. The ASM stores only the current 2-D-plus-time field ($365 \times 365 \times 425 \approx 57\,\mathrm{M}$ points) together with precomputed frequency-domain operators of the same shape, for a total of approximately 1.6 GB, a factor of nine reduction. For the present benchmark the FDTD completes in 1477 s (multi-GPU) while the ASM requires 516 s (single GPU), a 2.9× speed-up despite using only one GPU. The ASM wall time is dominated by its 260 sequential FFT-based propagation steps and could be reduced further with a coarser axial step.

6 Discussion

The results establish that the angular spectrum method, extended with phase-and-amplitude screens and plane-by-plane source injection, can reproduce FDTD-quality focal fields at a fraction of the computational cost for both transcranial and therapeutic ultrasound configurations.

The diffraction benchmarks confirm that the propagator correctly handles both unfocused and geometrically focused sources, matching analytical Rayleigh–Fresnel and Airy solutions with RMS errors of 0.086 (on-axis) and 0.002 (focal plane). The Strang+KT splitting achieves second-order convergence (rate ≈ 2) in the strongly nonlinear regime, compared with first-order for the sequential and Rusanov-based schemes. The KT flux resolves shocks more sharply than Rusanov (L^2 error 0.036 vs. 0.044), and the adaptive CFL sub-cycling guarantees stability at all step sizes while preserving accuracy.

The three boundary improvements act cumulatively, reducing the periodic-FFT reflection ratio by 2.4× without distorting the interior field. The frequency-weighted damping preferentially targets low-frequency components that see fewer wavelengths in the boundary layer, while the Engquist–Majda super-absorbing correction selectively removes incoming wave energy.

In the transcranial benchmark, the phase-and-amplitude-screen ASM matches the FDTD focal-plane intensity with 1.1% RMS difference and predicts a through-skull insertion loss of 5.4 dB, consistent with measured values of 5 dB to 10 dB [12]. The focal depth differs by only 2.6% from the FDTD reference. The TIPS bowl validation extends this agreement to a curved-source geometry: the plane-by-plane ASM matches the FDTD focal depth to within 2.3% while requiring 9× less memory and 2.9× less wall time.

Several limitations should be noted. The ASM is a forward-only propagator: the phase-and-amplitude-screen model captures the dominant wavefront aberration and insertion loss but neglects reflection, mode conversion, and multiple scattering at acoustic interfaces. The remaining discrep-

ancy with FDTD in the transcranial benchmark is concentrated in the pre-focal region and near the skull surface where these backward-propagating effects are strongest. For the bowl transducer, the FDTD's multi-layer volumetric source couples differently from the ASM's single-layer planar injection, requiring an empirical amplitude calibration factor (here 1.31) that depends on the FDTD grid spacing and number of source layers. The super-absorbing boundary uses a first-order Engquist–Majda decomposition, which is exact only for normally-incident waves. A higher-order formulation [6] could improve performance for wide-angle beams. Finally, although the Strang+KT splitting achieves second-order convergence when the nonlinear operator is tested in isolation, the full coupled solver includes additional error contributions from diffraction, transverse discretization, and the split-step composition of three operators, so the observed global convergence rate remains below two.

7 Conclusion

Two extensions to the angular spectrum method have been presented that address practical barriers to its use in transcranial and therapeutic ultrasound. Phase-and-amplitude screens derived from CT data incorporate skull-induced aberration and frequency-dependent insertion loss into the FFT-based propagator, matching an FDTD reference to 1.1% RMS in focal-plane intensity and predicting a through-skull insertion loss of 5.4 dB, consistent with measured values. Plane-by-plane source injection enables accurate modeling of spherical-bowl transducers whose curvature depth exceeds the wavelength. For a clinical TIPS annular array, the method matches the FDTD focal depth to within 2.3% while requiring 9× less memory and 2.9× less wall time.

On the numerical side, the second-order Kurganov–Tadmor flux with Strang splitting achieves rate-2 convergence and, with adaptive CFL sub-cycling, remains stable at all tested step sizes. Three absorbing-boundary improvements reduce periodic-FFT reflections by 2.4× without distorting the interior field. The intensity-loss tracking provides a spatially resolved absorbed-energy map that accounts for the full harmonic content of the nonlinear wave, suitable for radiation-force and thermal dose calculations.

Together, these developments extend the practical reach of the angular spectrum method from homogeneous free-field propagation to heterogeneous transcranial paths and immersed curved-source geometries, at computational costs that are compatible with treatment planning and real-time aberration correction workflows.

The source code is available at `https://github.com/gfpinton/angularspectrum_python` under the LGPL v3 license.

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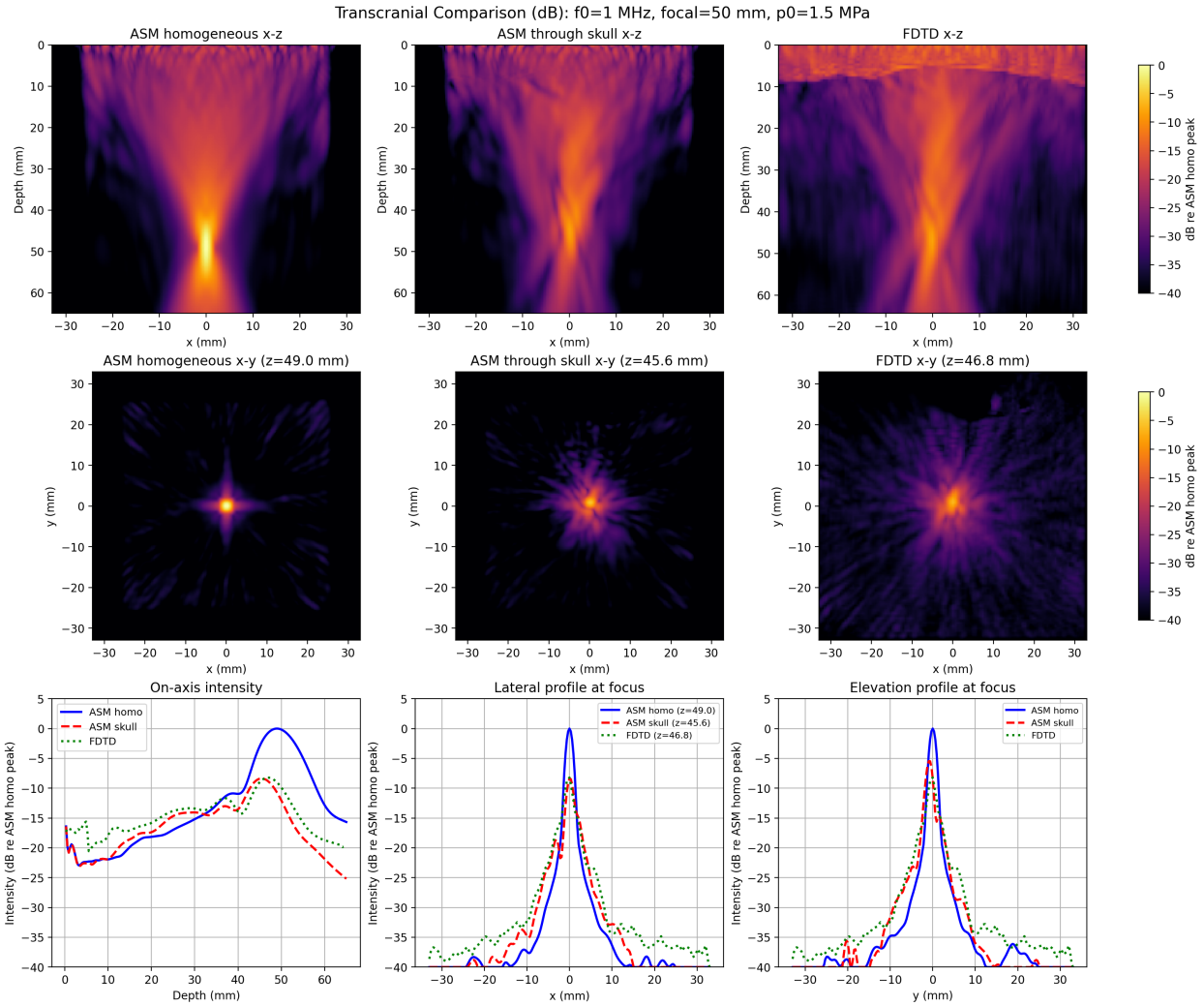

Figure 8: Transcranial comparison referenced to the homogeneous ASM field. Top row: axial (x–z) beam profiles for ASM homogeneous, ASM through skull, and FDTD. Middle row: corresponding focal-plane (x–y) intensity maps at each method's focal depth. Bottom row: on-axis, lateral (x), and elevational (y) profiles in dB. The FDTD intensity is weighted by its recorded temporal sampling interval before normalization so the dB offsets are consistent with the ASM time-integrated intensity.

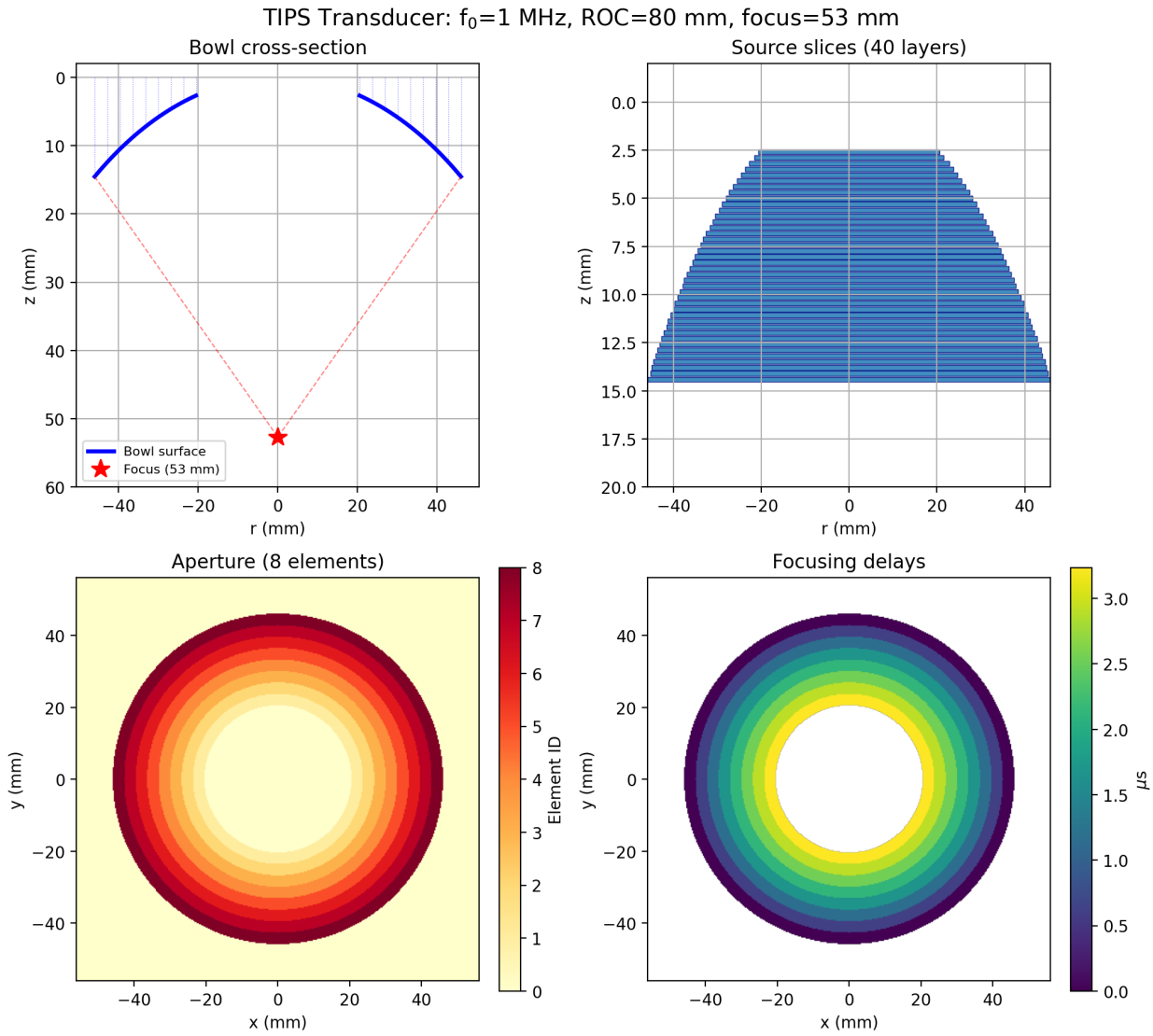

Figure 9: TIPS bowl transducer setup. Top left: bowl cross-section ($R = 80\,\mathrm{mm}$) with element boundaries and focal lines. Top right: plane-by-plane source slices spanning the 14.6 mm bowl depth. Bottom left: 8-element annular aperture. Bottom right: per-element focusing delays.

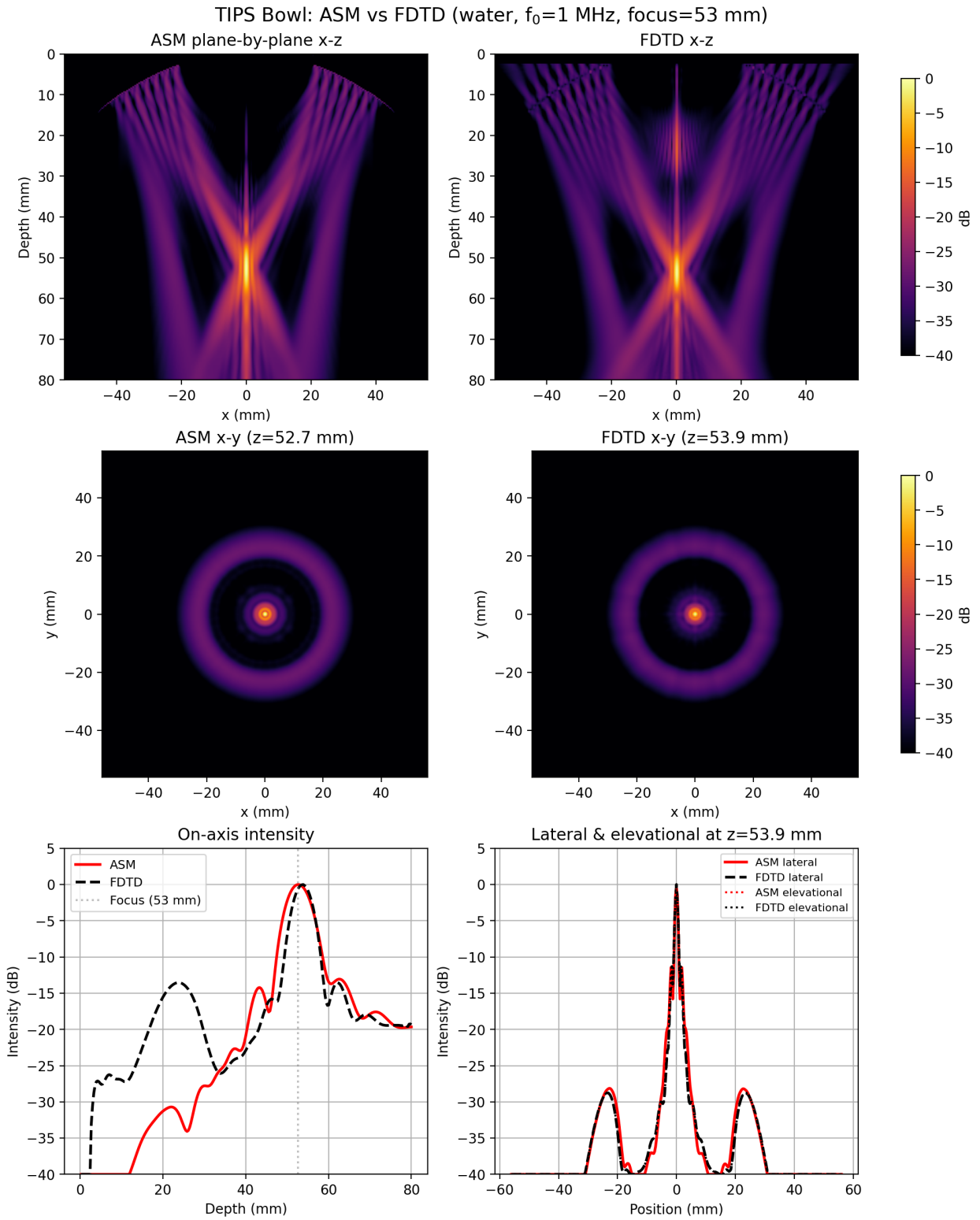

Figure 10: TIPS bowl comparison: ASM plane-by-plane vs FDTD in water. Top row: axial (x–z) beam profiles. Middle row: focal-plane (x–y) intensity maps at each method's focal depth. Bottom left: on-axis intensity. Bottom right: lateral and elevational profiles at the FDTD focal depth. Each method is normalized to its own peak so the beam shapes can be compared directly. The ASM focal depth (52.7 mm) matches the FDTD (53.9 mm) to within 2.3%.